

LECTURE 3:

AGENT TYPES

Introduction to Multi-Agent Systems (MESIIA, MIA)

Department of Computer Science and Mathematics

Rovira i Virgili University

Fall 2019

Overview

- Agent **typology**
 - Description of some types of agents
 - **Collaborative** agents (intro)
 - **Interface** agents
 - **Information** agents
 - **Heterogeneous systems**
 - **Agentification** mechanisms
-

Agent properties

[Previous lecture]

- Autonomy
- Reactivity
- Proactiveness
- Communication
- Reasoning
- Learning
- Mobility
- ...

How to classify agents?

- There are many different classifications of agents, depending on the employed criteria
 - Researchers from BT Lab proposed a classification that depends on the **properties that are emphasized** in a particular agent
 - The classification is not an exact division !!!
-

Agent classification (BT Lab) (I)

- Collaborative Agents
 - Communication, autonomy, reasoning
- Interface Agents
 - Learning, autonomy, proactivity
- Mobile Agents [recall lecture 2]
 - Mobility
- Internet / Information agents
 - Learning, autonomy, proactivity, mobility

Agent classification (BT Lab) (II)

- **Reactive agents** [recall lecture 2]
 - Reactivity
- **Hybrid agents** [recall lecture 2]
 - Reactivity + reasoning (deliberative)
- **Heterogeneous systems**
 - Multi-agent systems with different types of agents

Collaborative Agents (I)

- Emphasize *autonomy*, as well as *communication* and *cooperation* with other agents
- Typically operate in open multi-agent environments => *multi-agent systems* (MAS)
- *Negotiate* with their peers to reach mutually acceptable agreements during cooperative problem solving



Collaborative Agents (II)

- They normally have very limited **learning** capabilities
- Collaborative agents are usually *deliberative* agents (e.g. based on the BDI model), with some **reasoning** capabilities
 - Reactive agents can hardly communicate and collaborate (only through actions that modify the common environment)

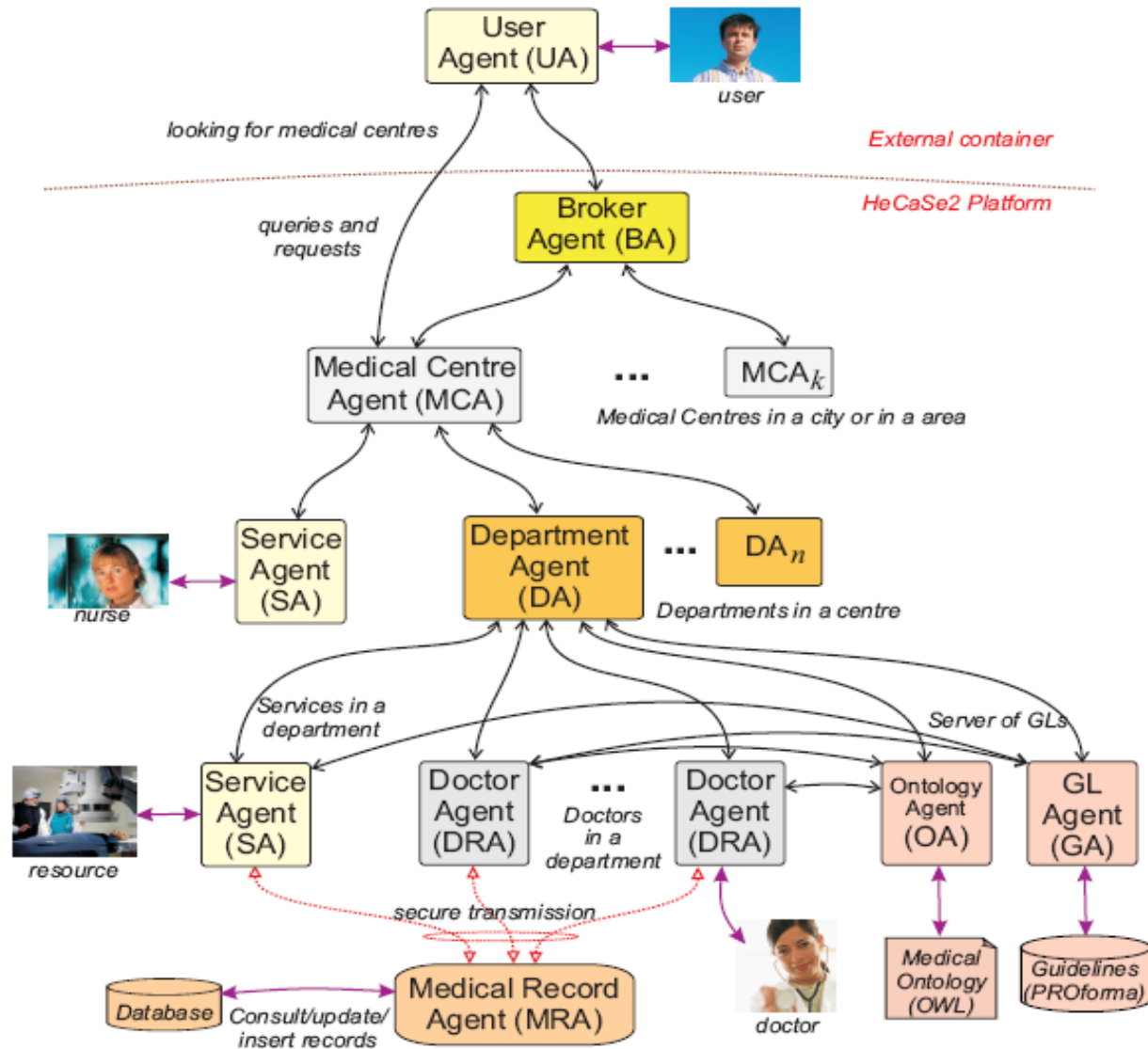
[MAS = second part of the course]

Collaborative Agents: Rationale

- To solve problems that are too large for a single centralised agent
 - To create a system that functions beyond the capabilities of any of its members
 - To allow for the interconnection and inter-operation of existing legacy systems
 - To provide solutions to inherently distributed problems
-

Collaborative Agents: Applications

- Provide solutions to **physically distributed problems**
 - air-traffic control, management of a team of robots
- Provide solutions to problems with **distributed data sources**
 - different offices of a multi-national business
- Provide solutions that need **distributed expertise**
 - health care provision (family doctors, nurses, specialists, laboratory analysis, ...)



HeCaSe2 Architecture

Interface Agents (I)

- Emphasise *autonomy* and *learning* in order to perform tasks for their owners
 - Support and provide *proactive* assistance to a human that is using a particular application or solving a certain problem
 - Anticipate user needs
 - Make suggestions
 - Provide advice
 - ... without explicit user requests
-

Interface Agents (II)

- Limited **cooperation** with other agents
 - Limited **reasoning** and **planning** capabilities
 - Interface agent = **personal assistant** = personal digital assistant = personal agent
 - Kind of “secretary” that helps the user in his work environment
-

Personal Assistant metaphor (I)

- Initially, a personal assistant is not very familiar with the habits and preferences of his employer
 - ❑ It may not be very helpful
 - ❑ It may even give extra work !

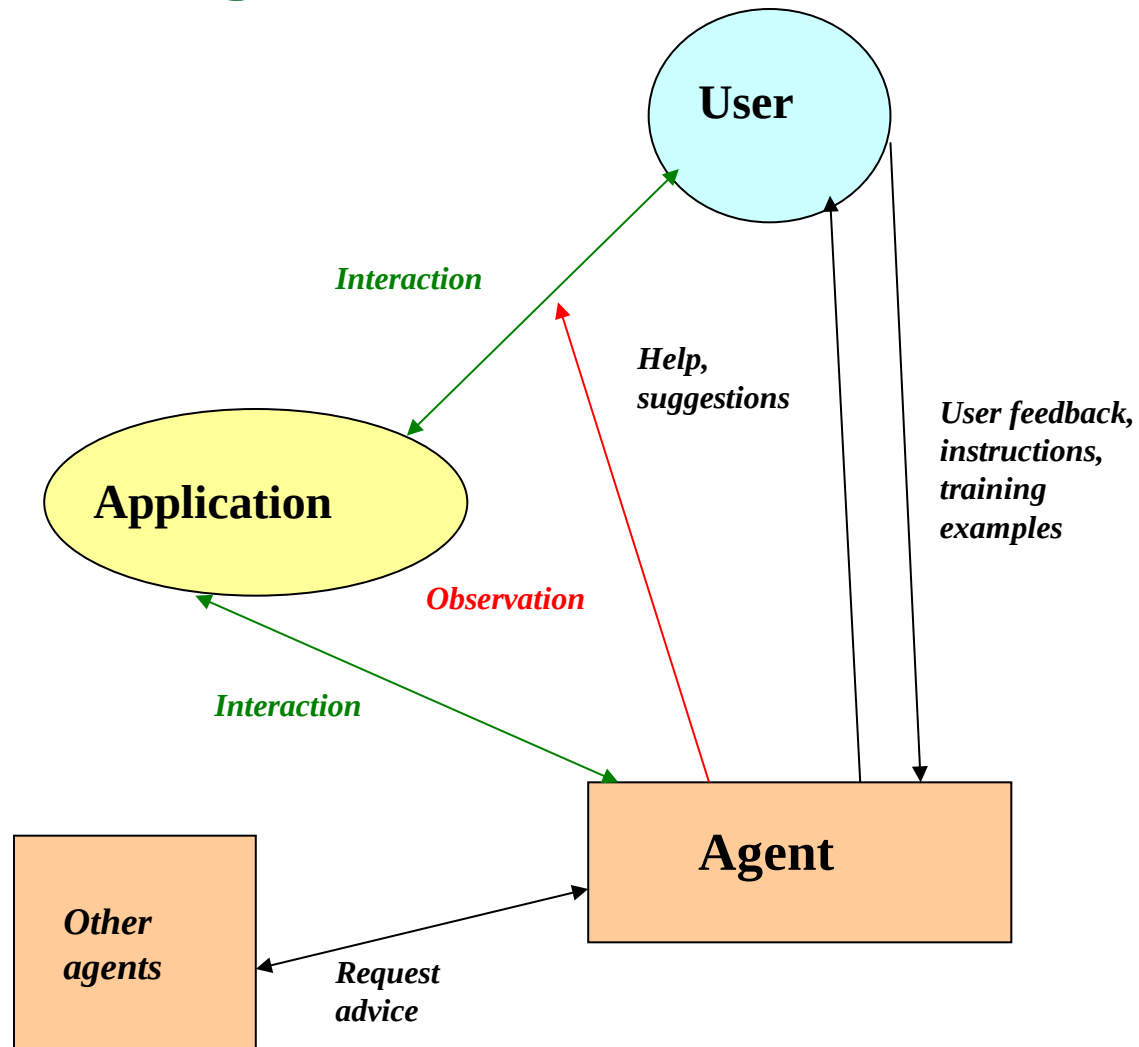
Personal Assistant metaphor (II)

- With every experience, the assistant *learns* by
 - watching how the employer performs tasks
 - receiving instructions from the employer
 - learning from other more experienced assistants
 - Gradually, more tasks that were initially directly performed by the employer can be taken care of by the assistant
-

Problems in building interface agents

- **Competence:** How does an agent acquire the knowledge it needs to decide:
 - **when** to help the user
 - **what** to help the user with
 - **how** to help
 - **Trust:** How can we guarantee the user feels comfortable delegating tasks to an agent?
-

Interface Agents



Interface Agents: Rationale (I)

- Less work for the end user and for the application developer
 - Automation of routine activities
 - Automation of tasks that would take a long time to a human user
- The agent can adapt, over time, to its user's preferences and habits
 - Learn automatically user profiles, adapted to the user needs
 - The profile of a user can change dynamically

Interface Agents: Rationale (II)

- Know-how among different users in a community may be shared
 - Communication and (limited) cooperation between interface agents of different users
- Use in applications with repetitive behaviour
 - Even if the behaviour is different for each user

Interface Agents: Applications

- Mail management
 - Scheduling meetings
 - Internet browsing
 - News filtering agent
 - Buying/selling on behalf of the user
-

Example 1: mail management assistant

- **Delete** uninteresting or potentially harmful messages
 - **Prioritize** the messages according to their relevance
 - **Sort** the incoming messages in the appropriate folders
 - **Warn** the user when a very important message arrives
 - **Forward** a message to another user
-

Example 2: meeting scheduler

- Learn user preferences on different kinds of meetings
 - Make a meeting proposal
 - Accept a meeting proposal
 - Reject a meeting proposal
 - Reschedule a previously agreed meeting
 - Negotiate a meeting time with other agents
-

Example 3: Web browsing - Letizia

- Agent developed at MIT (Lieberman, 1997)
 - Helps the user when he is browsing the Web to look for information
 - The agent is actually embedded in the browser
 - Letizia **monitors the behaviour** of the user while browsing through Internet
 - Observes the user's actions and makes some **reasoning** on them
 - Gradual **automatic discovery** of the **user interests**
-

Some heuristics in Letizia (I)

- If the user saves a **bookmark**, that shows interest in the content of the current page
- If the user **follows a link**, that shows a potential interest in the content of the linked page
- If he **comes back from a page**, that probably shows that the page was not interesting after all

Some heuristics in Letizia (II)

- If the user **follows a link in the middle of a Web page**, that probably indicates that he was not interested in the previous links
 - If a user **spends a long time** in a Web page, or **returns** often to a certain page, he is probably interested in its content
 - If the user **makes an explicit query** to a Web search engine, the used terms describe information in which he is interested
-

User profile

- Letizia uses the previous heuristics to discover pages in which the user is interested
 - Analyzing the content of these pages, the agent may discover the terms that appear with **more frequency**
 - Those terms can be used to represent the **user profile**
[agents, multi-agent systems, tennis, Davis Cup]
-

Relevant properties

- The profile is made **automatically**
 - Without bothering/interrupting the user
 - Without explicit instructions or information from the user
 - The profile is **updated** automatically, when the interests of the user change
 - Example: new job, new family status
-

Web browsing

- Standard web browsing
 - ❑ Basically *depth-first search*
 - ❑ Follow a link from the current page, then a link from that page, etc.
 - ❑ Sometimes the user can go back to the previous page to try another link, or select a bookmark

The screenshot illustrates a web browsing session. At the top is a menu with the following items:

Go	Bookmarks	Options	Directory
Back			⌘[
Forward			⌘]
Home			
Stop Loading			⌘.
✓ Letizia			⌘0
Papers by Henry Lieberman			⌘1
Henry Lieberman's Home Page			⌘2
Henry Lieberman's Bookmarks			⌘3

Below the menu are three web pages:

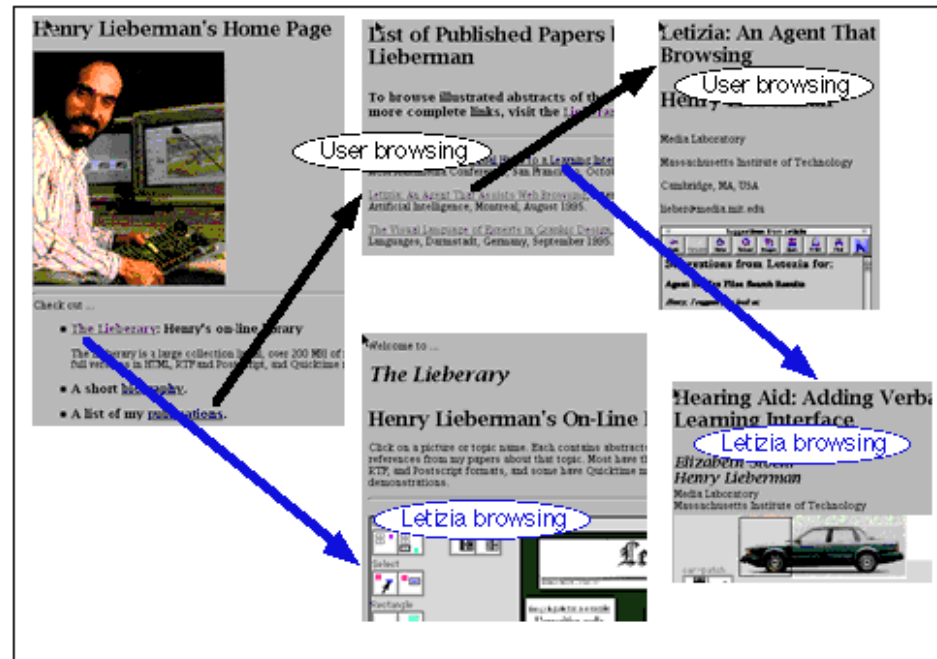
- Henry Lieberman's Home Page**: Features a photo of Henry Lieberman and a list of links: "The Lieberman: Henry's on-line library", "A short biography", and "A list of my publications".
- List of Published Papers by Lieberman**: Contains text about browsing illustrated abstracts and links to "Letizia: An Agent That Assists Web Browsing" and "The Visual Language of Events in Graphical Systems".
- Letizia: An Agent That Browsing**: Displays information about the Media Laboratory at MIT and suggestions from Letizia.

Two black arrows labeled "User browsing" indicate the navigation path: one from the "Henry Lieberman's Home Page" to the "List of Published Papers by Lieberman", and another from the "List of Published Papers by Lieberman" to the "Letizia: An Agent That Browsing" page.

It may be difficult for the user to find something, even if it is only 3-4 clicks away

Letizia web browsing

- Letizia conducts a *concurrent breadth-first* search rooted from the user's current position

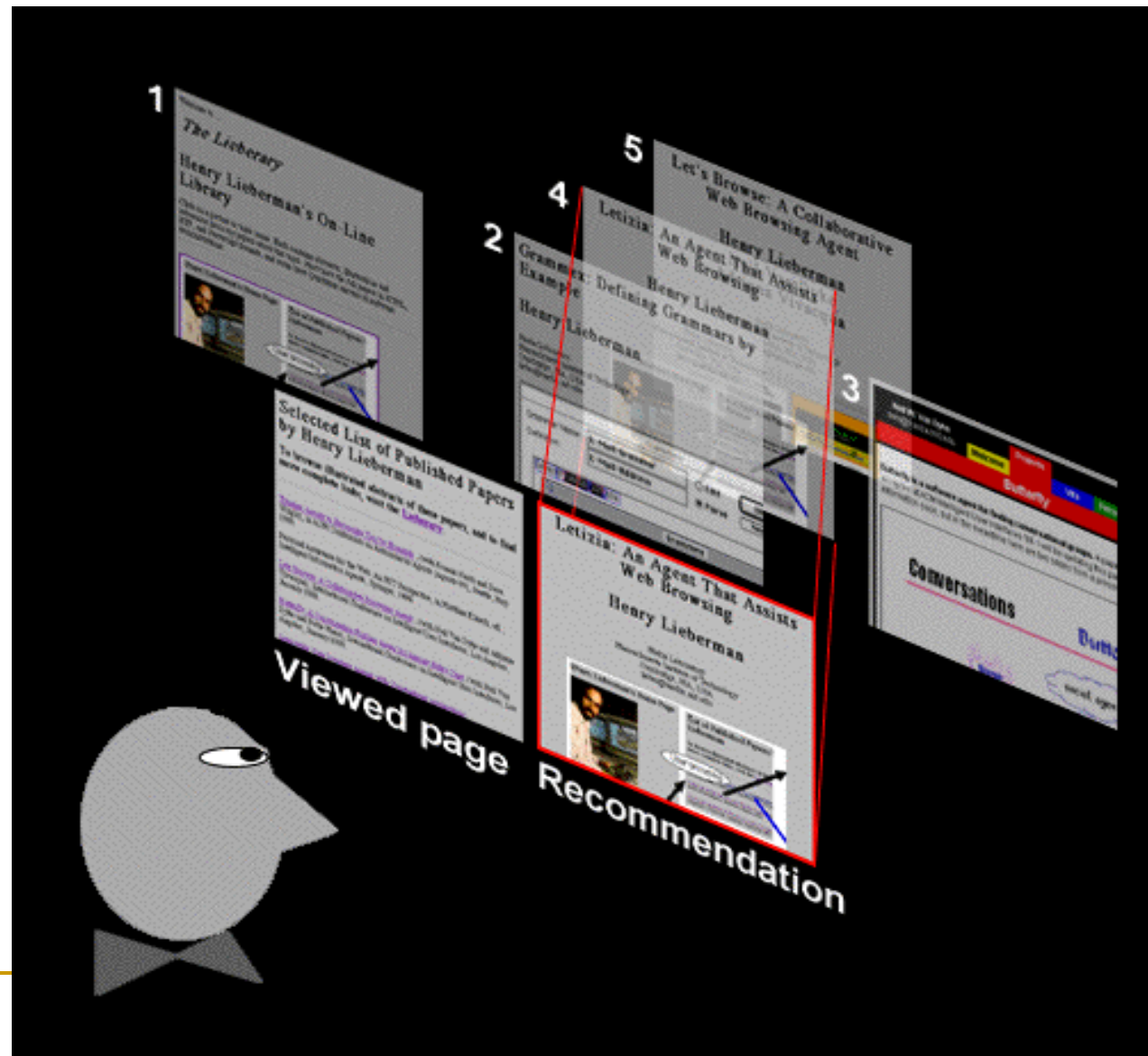


Automatic link suggestions

- While the user reads a page, Letizia **analyzes the pages linked to it**, in a depth-first fashion
 - It can discover which of those pages (at several levels of distance) can be **interesting** to the user
 - Content related to the user interests contained in his **profile**
 - It can also discover **dead links**
 - Letizia can **suggest** to the user links to be followed in the current page
-

Letizia recommendation

Video: short demo



Problems of personal assistants

■ Slow learning curve

- ❑ Agents require many examples before they can make accurate predictions (especially if they cannot be directly trained)
- ❑ No useful assistance during the learning process

■ Learning from scratch

- ❑ Each agent has to learn on its own, even if there is a bunch of agents dealing with a team of people with similar interests

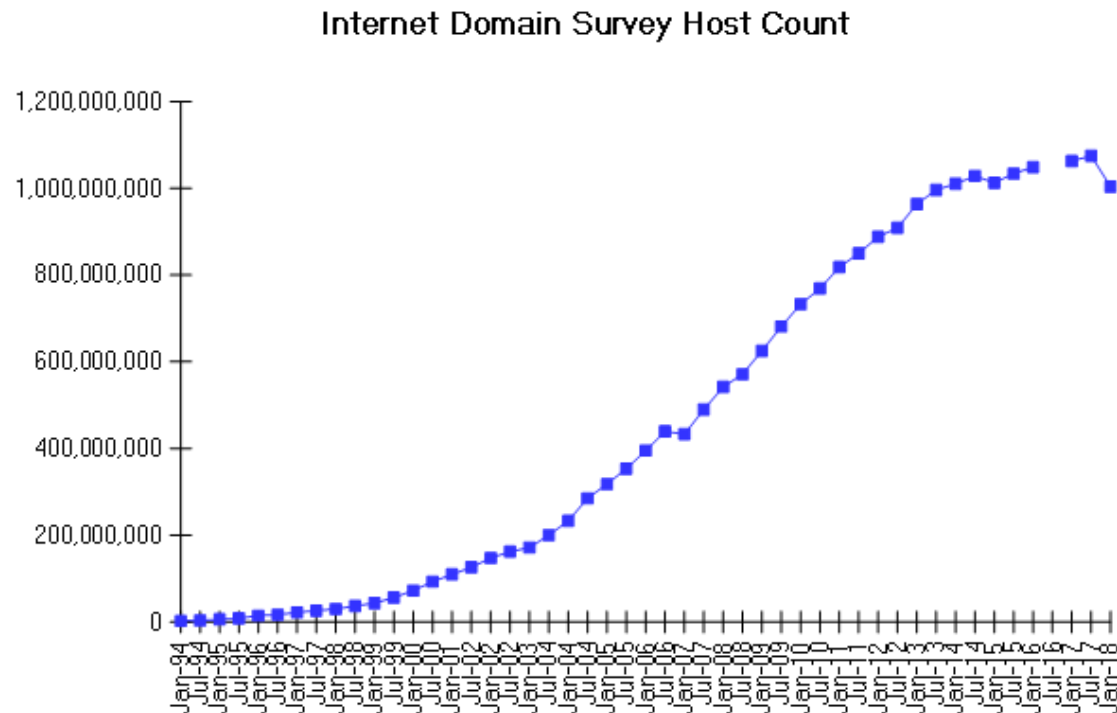
■ Difficulty to adapt to completely new situations

Information Agents

- Software agents that manage the access to multiple, **heterogeneous** and geographically **distributed information sources**.
 - Information agents = *Internet agents*
 - Main task: **proactive acquisition**, **mediation** and **maintenance** of **relevant** information for a user/other agents
-

Interest of Information Agents (I)

- Need of tools to manage the **information explosion** of the WWW
 - Impossible manual management



Source: Internet Systems Consortium (www.isc.org)

Interest of Information Agents (II)

- Commercial benefits

- *Proactive, dynamic, adaptive and cooperative*

- WWW browser-embedded information manager

- User benefits

- Time and effort to access and analyze data

- Improve productivity (more time, better data)

Tasks of information agents (I)

- Information acquisition and management
 - Provide transparent **access** to different information sources
 - **Retrieve**, **extract**, **analyze**, **summarize** and **filter** data
 - **Monitor** information sources
 - **Update** relevant information on behalf of the user
 - Examples: DBs, web pages, purchase of information from providers on electronic marketplaces, ...
-

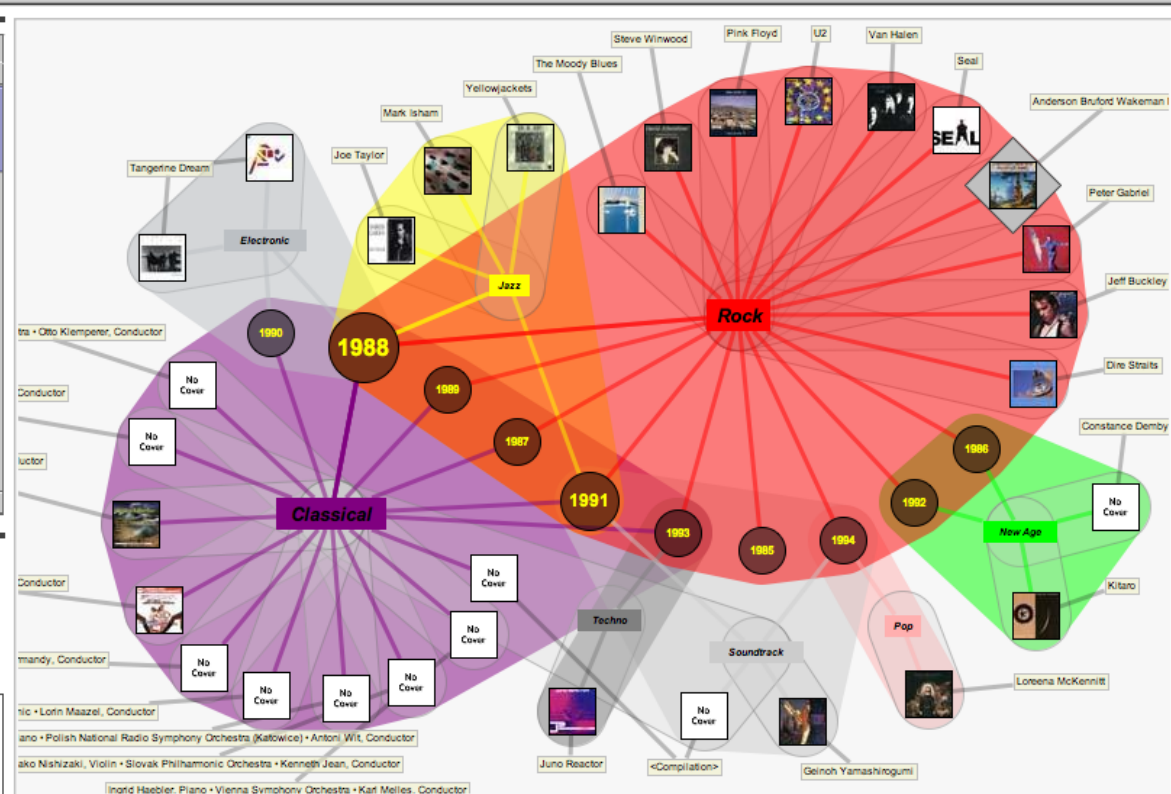
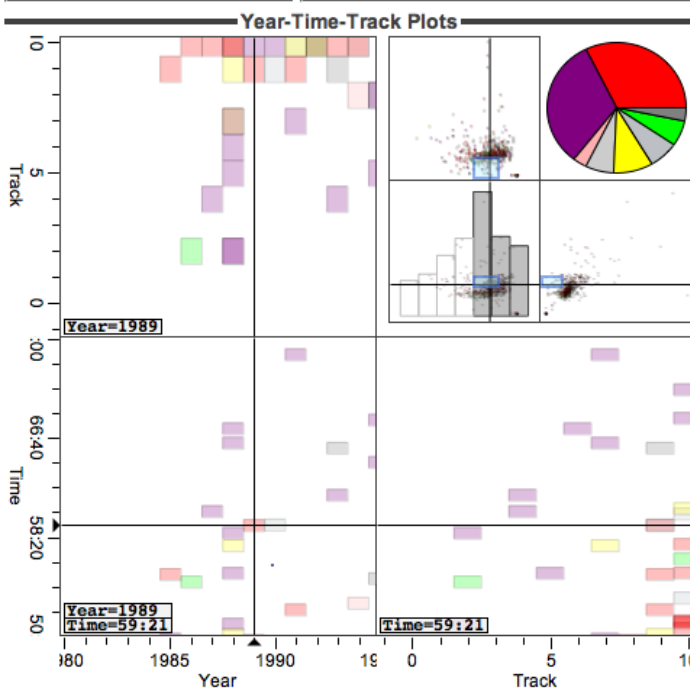
Tasks of information agents (II)

- Information synthesis and presentation
 - Fuse, merge heterogeneous data
 - Handle conflicts, contradictions, repetitions
 - Provide unified, multi-dimensional views on relevant information to the user
 - Not just a mere list of data from different sources
-

Tasks of information agents (III)

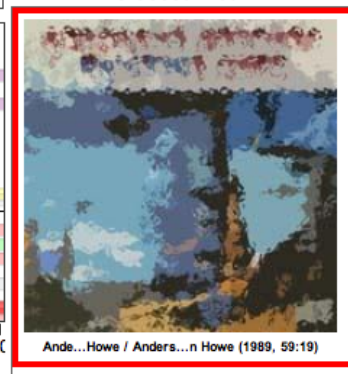
- Intelligent user assistance
 - Learn automatically user preferences
 - Adapt dynamically to changes in user preferences
 - Personalised presentation of information
 - Use of intelligent user interfaces
 - Sometimes with believable, life-like characters
-

Decades			Genres		
Decade	Detail	Count	Genre	Detail	Count
2000s		0 of 39	Rock		10 of 214
1990s		14 of 251	Classical		10 of 58
1980s		17 of 178	Soundtrack		2 of 50
1970s		0 of 54	New Age		2 of 49
1960s		0 of 12	Alternative		0 of 45



Nodes	
☒ Artists	
☒ Genres	
☒ Names	
☒ Years	
Edges	
☐ Artist-Genre	
☒ Artist-Name	
☐ Artist-Year	
☒ Genre-Name	
☒ Genre-Year	
☐ Name-Year	
Packs	
☒ A-G	☐ G-A
☒ A-N	☐ N-A
☐ A-Y	☐ Y-A
☒ G-N	☐ N-G
☒ G-Y	☐ Y-G
☐ N-Y	☐ Y-N
Settings	
☐ Label Names	
☒ Obscure Covers	
☐ Plot Distortion	

Closest Album



Albums

Artist	Name	Co...	Year	Genre	Tracks	Time
Anderson Bruford Wakeman Howe	Anderson Bruford Wakeman Howe		1989	Rock	9	59:19
Tangerine Dream	Melrose		1990	Electronic	9	59:22
Ingrid Haebler, Piano • Vienna ...	Mozart: Piano Concerti Nos. 19 & 20		1988	Classical	2	58:40
Mark Isham	Castalia		1988	Jazz	7	57:37
<Compilation>	Robin Hood: Prince Of Thieves		1991	Soundtrack	10	60:18
Detroit Symphony Orchestra • N...	Tchaikovsky: 1812 • Romeo And Juliet • March...		1987	Classical	4	60:30
Yellowjackets	Greenhouse		1991	Jazz	10	60:46

General

Adaptation ☒ 4769
Climate Change ☒ 307k
Climate Policy ☒ 3423
Climate Science ☒ 4958
Mitigation ☒ 3606

Renewable Energy

Biomass ☐ 21063
Geothermal ☐ 9833
Hydro ☐ 8183
Solar ☐ 162k
Wind ☐ 28706

World Summits

Lima (COP20) ☐ 239
Paris (COP21) ☐ 62604
UNCSD (Rio+20) ☐ 535
Warsaw (COP19) ☐ 98

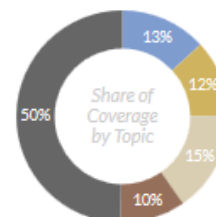
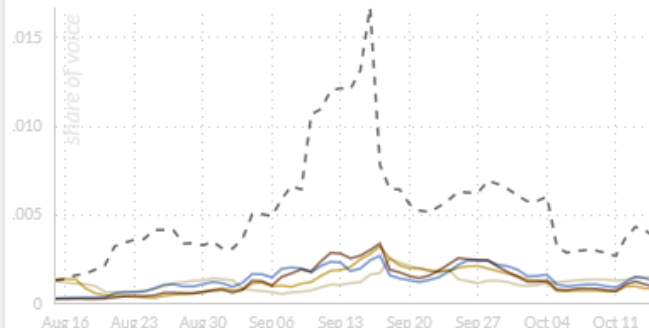
Events and Campaigns

Earth Day ☐ 563
Earth Hour ☐ 1995
IPCC Report ☐ 5635

Associations Search History

deforestation Brazil 17965

germany ☐ 20249
climate change ☐ 307k
action ☐ 74075
rt greenpeace ☐ 12844
world ☐ 139k
amazon ☐ 8760
climatechange ☐ 57166
energy ☐ 1632k
palmoil ☐ 10006
rating ☐ 21771
borneo ☐ 1268
major ☐ 45852
end ☐ 63114
global ☐ 195k



08/22 OIC



and revolutionising the way they are operated. Countries such as Brazil and China and those in Africa are at t...
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08/22 The Road to Economic Growth



and revolutionising the way they are operated. Countries such as Brazil and China and those in Africa are at t...
blog.agcocorp.com

09/30 Awards and Reviews



really shocked us. O Estado de São Paulo - Daily Newspaper - Brazil. "ION performance is outstanding. (...) F...
nvidia.com

10/03 summer weight down comforter - Search Results



Gradis hardwood, grown in 100 % well managed forests in Brazil, certified by the FSC (Forest Stewardship C...
Int.com

09/11 Good Manufacturing Practices - GMP



installed new processing equipment in our facilities in Africa, Brazil, Canada, Italy, and the United States. SO...
universalcorp.com

09/26 Quest Diagnostics of Brazil - Contact Us



of Brazil is proud to have a relationship with the main hospitals and laboratories across the country. Together...
questdiagnostics.com

09/21 Diversity and Inclusion



around the world. We have a significant presence in Australia, Brazil, Canada, China, Germany, the Netherlan...
ashland.com

10/07 O-I - Investors - Company News Release



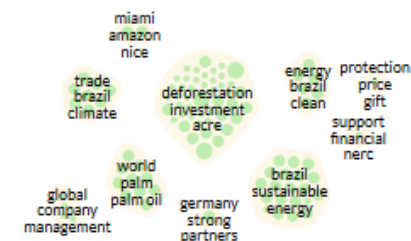
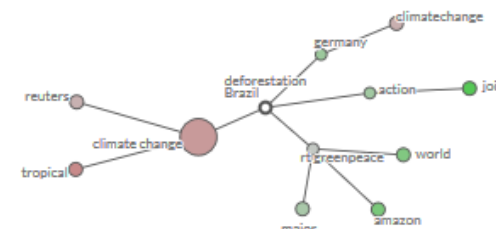
s production," said Johnston. Yaguara is currently available in Brazil, North America, and the U.K. Photo: (fro...
investors.o-i.com

10/07 Improving Efficiency - Goodyear Corporate

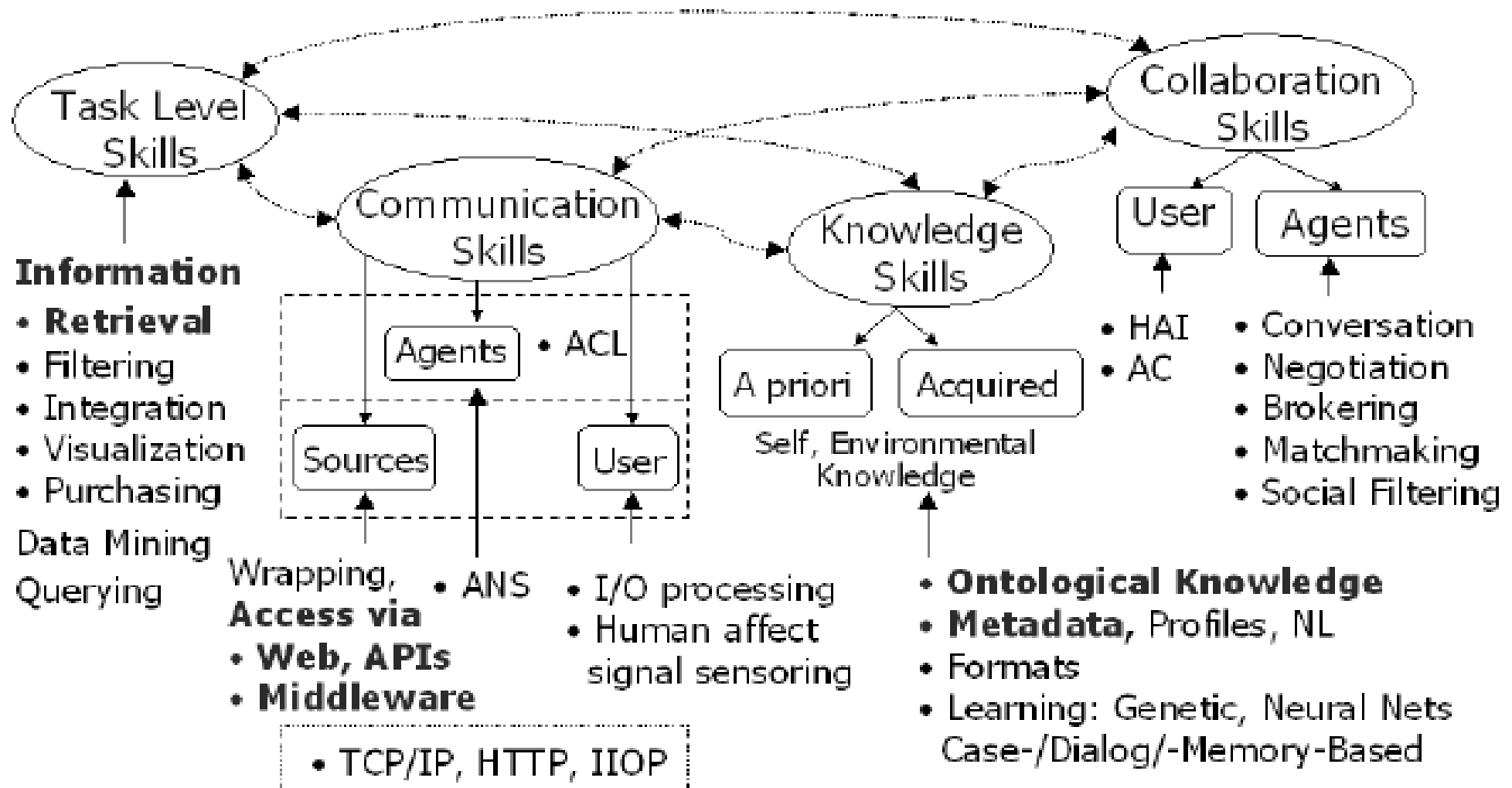


as part of a five-year plan. Goodyear's facility in Americana, Brazil, installed more efficient fan motors in the c...
corporate.goodyear.com

actonclimate alternative amazon animal area auction borneo
brazilian business cheap china climateaction climatechange
common cop cost credit damage ecological end environment
environmental food free future germany htt impact india
indonesia investment joint land major mexico nature palmoil
peru plan production project rating report reuters rio satellite
social study supply sustainability tower trade trouble water wwf

[Video](#)[Web page](#)[Another web page](#)

Basic skills of an information agent



Information retrieval

- Classical view: management of huge volumes of data in **centralised, static databases**
 - Standard **conversational** paradigm
 - User makes a question
 - System returns the subset of documents that are considered **relevant** to the query
 - User evaluates the returned information items
 - User refines the initial question
-

Information retrieval methods

■ Input

- Set of documents
 - Domain corpus, or the Web
- Query

■ Output

- Subset of the documents relevant to the query

■ Requirements

- Way of representing the **info.** of a document
 - Way to measure the **relevance** of a document with respect to a query
-

Vector space model

- N text documents, containing t terms
- Each document is represented by a t -dimensional vector
 - Each component of the vector represents the weight of the term in the document

$$d_j = (w_{1j}, w_{2j}, \dots, w_{tj}) \quad j \text{ in } 1..N$$

Assigning weights to terms

- **TFIDF: *Term Frequency Inverse Document Frequency***

- The weight of a term in a document measures the relevance of that term to represent the document

$$w_{ij} = \text{tf}_{ij} * \log (N / \text{df}_i)$$

tf_{ij} = number of appearances of term i in document j

N = total number of documents

df_i = number of documents in which term i appears

[e.g. a term that appears in all docs. has weight 0]

Similarity

- The similarity between two documents (or a query q and a document d_j) is computed with the **cosine of the angle** formed by the two term vectors (scalar product divided by the product of the norms)

$$\text{sim}(d_j, q) = \frac{\vec{d}_j \bullet \vec{q}}{|\vec{d}_j| \times |\vec{q}|} = \frac{\sum_{i=1}^t w_{i,j} \cdot w_{i,q}}{\sqrt{\sum_{i=1}^t w_{i,j}^2} \cdot \sqrt{\sum_{i=1}^t w_{i,q}^2}}$$

Example

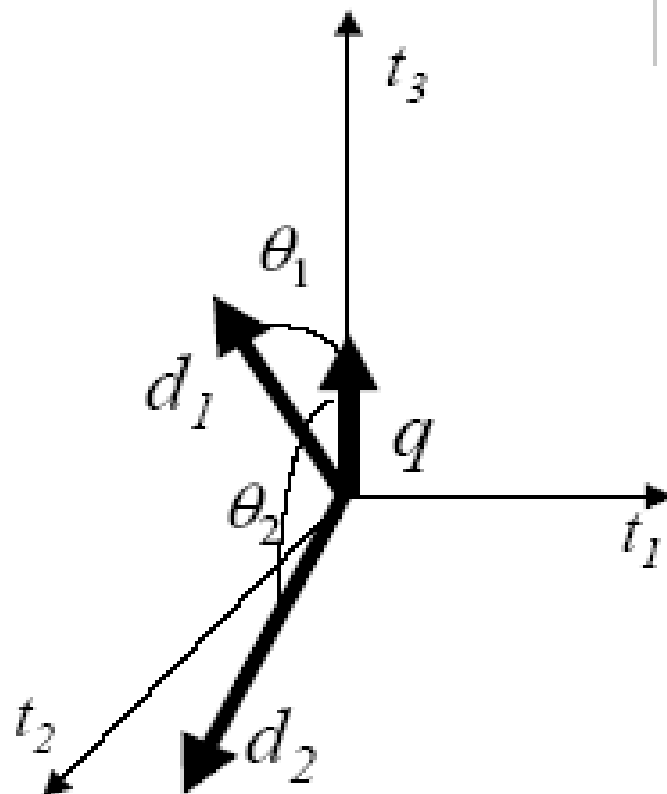
Query $q = (0, 0, 2)$

Document $d_1 = (2, 3, 5)$

$$\text{Sim}(d_1, q) = 10 / \sqrt{(4+9+25)(0+0+4)} = 0.81$$

Document $d_2 = (3, 7, 1)$

$$\text{Sim}(d_2, q) = 2 / \sqrt{(9+49+1)(0+0+4)} = 0.13$$



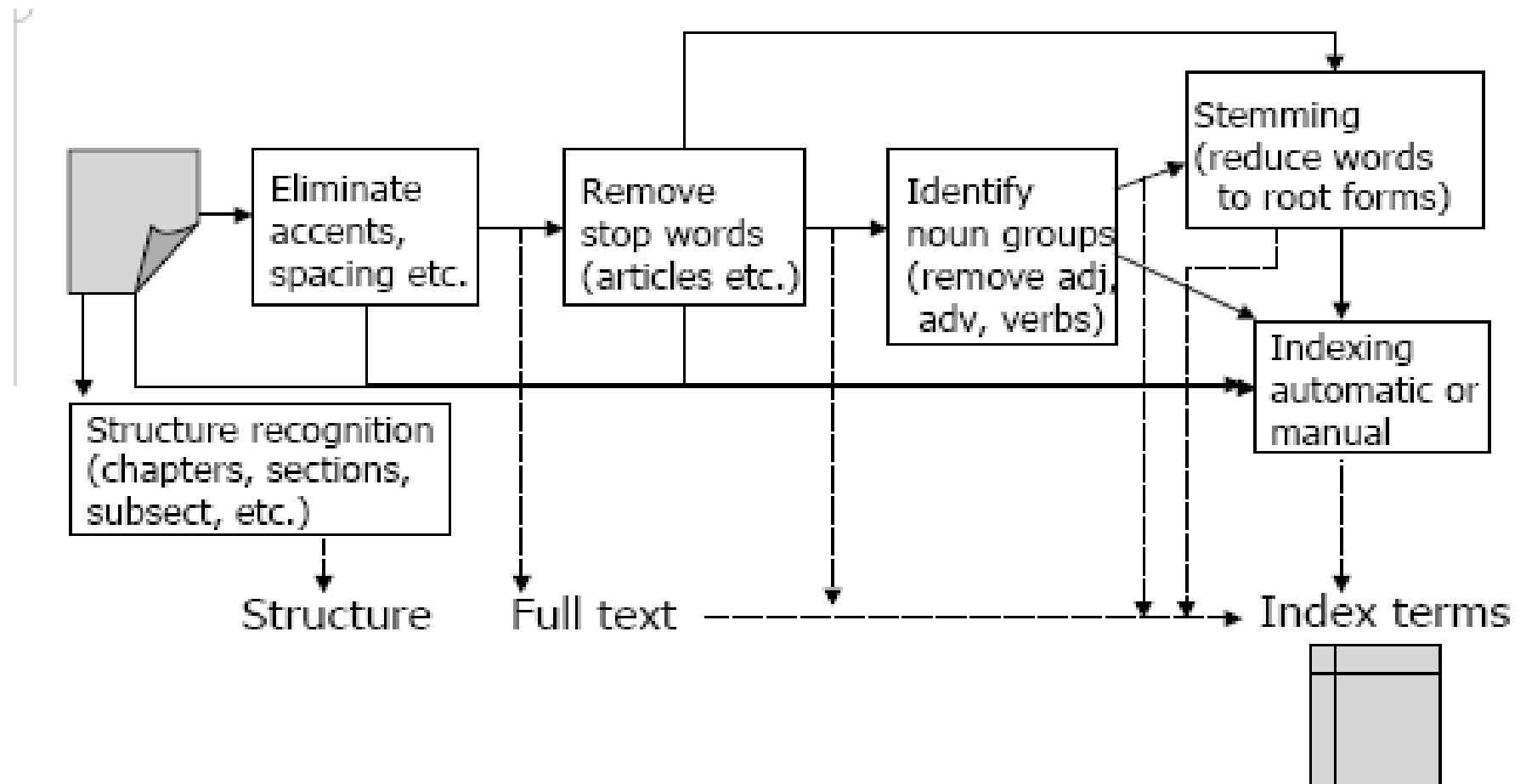
Preprocess: Stopwords

- Filtering out words with **very low discrimination** values, to reduce the size of vector terms
 - Highly frequent words without individual meaning
 - Articles
 - Prepositions
 - Conjunctions
 - Adverbs
 - ...
-

Preprocess: Stemming

- *Stemming* is the process of reducing a given word to its **grammatical root**
- **Group words** with common semantics
- **Reduce size** of vector terms
- Add **flexibility** to user queries
- **Stem** = portion of a word left after removal of its affixes (i.e. prefixes, suffixes)
 - Computing / computer / computers / computation / compute / computes / uncomputable => stem: **comput**

Preprocess: Index building



Document 1

The bright blue butterfly hangs on the breeze.

Document 2

It's best to forget the great sky and to retire from every wind.

Document 3

Under blue sky, in bright sunlight, one need not search around.

Stopword list

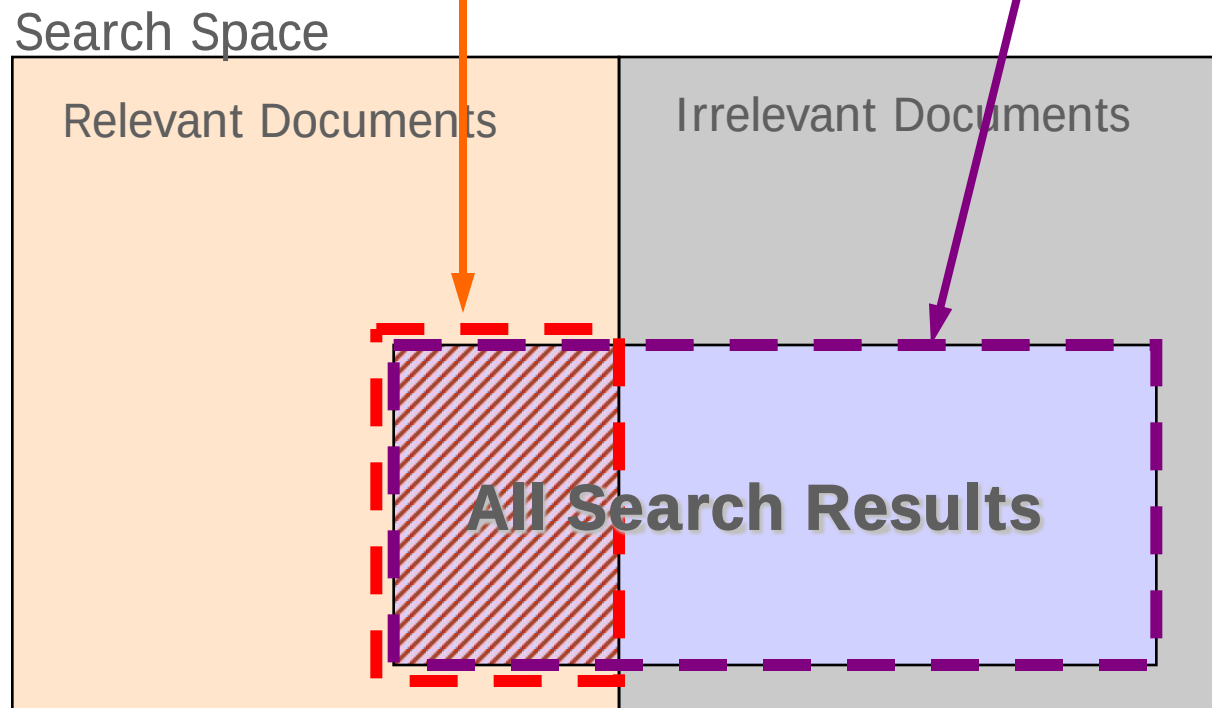
a
and
around
every
for
from
in
is
it
not
on
one
the
to
under

Inverted index

ID	Term	Document
1	best	2
2	blue	1, 3
3	bright	1, 3
4	butterfly	1
5	breeze	1
6	forget	2
7	great	2
8	hangs	1
9	need	3
10	retire	2
11	search	3
12	sky	2, 3
13	wind	2

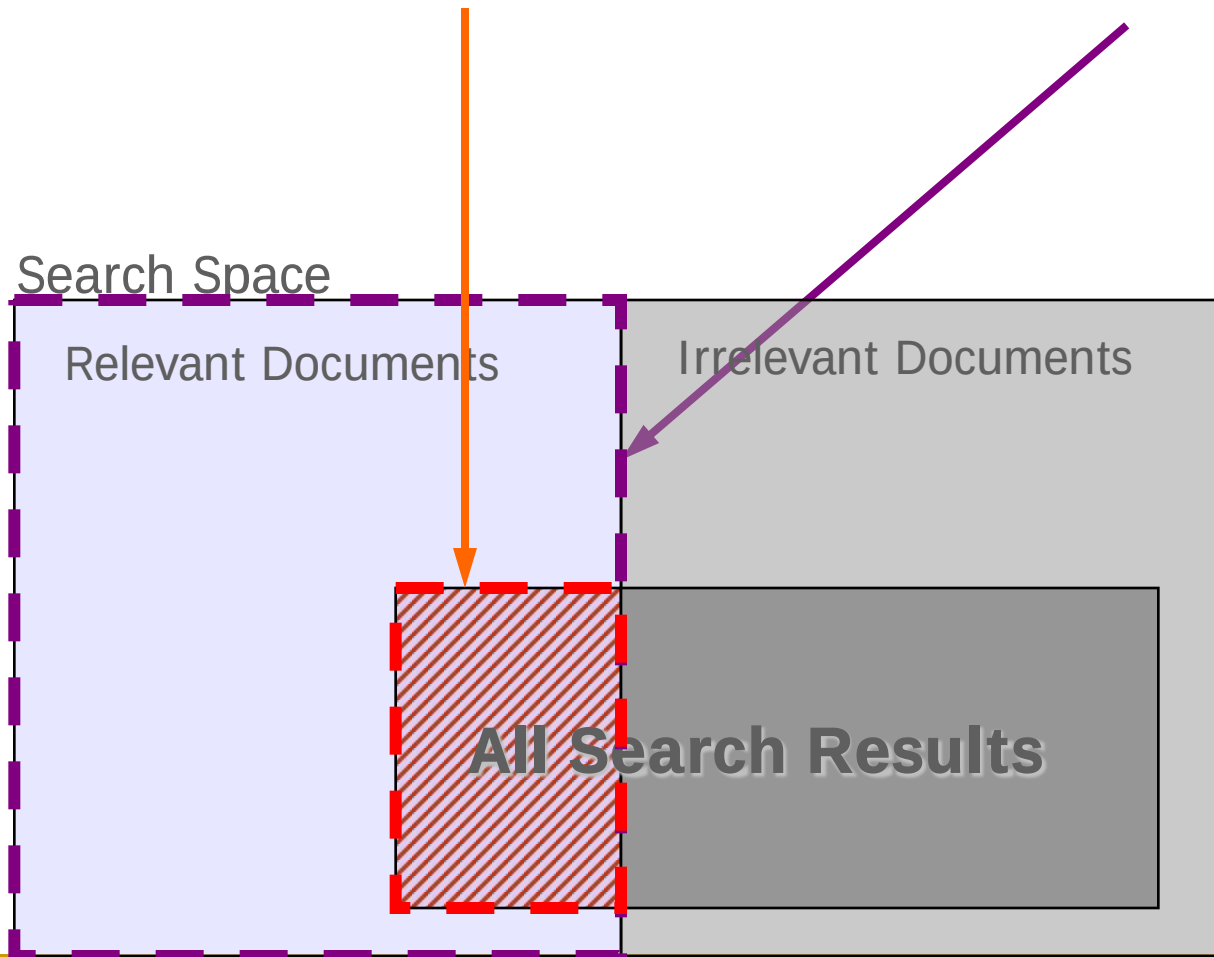
IR measures: Precision

- Precision: $\frac{\text{Relevant Search Results}}{\text{All Search Results}}$



IR measures: Recall

■ **Recall:** $\frac{\text{Relevant Search Results}}{\text{All Relevant Documents}}$



Precision vs recall

■ Precision

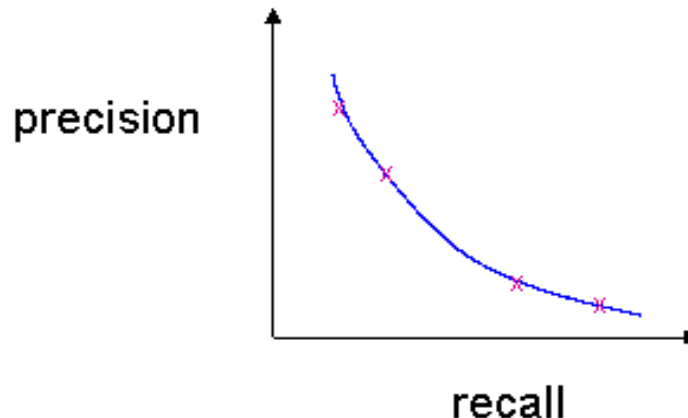
- ❑ Of the documents that have been retrieved, how many of them are good?
- ❑ **High precision:** *the system does not select bad documents*

■ Recall

- ❑ Of all the good documents, how many of them have been retrieved?
 - ❑ **High recall:** *the system does not discard good documents*
 - ❑ *Normally not computable in open environments like the Web*
-

Tradeoff between precision and recall

- ❑ System returns 1 very good document:
 - 100% precision, very low recall
- ❑ System returns all documents:
 - 100% recall, very low precision



Searching for information on the Web

- Option 1: Web search engines
 - Process text of web pages (stopwords, stemming)
 - Compute **terms vector** for each page
 - Store, for each page, the terms that have a weight over a certain threshold
-

Web search engines indexes

- ❑ Build an **inverted index** (term $\rightarrow$ pages), to be used in the queries
- ❑ Continuous Web crawling, analysis of web pages and updating of indexes
 - This approach requires a **very high** level of resources and bandwidth

Example: Google – see Wired article

- Around 24 data centres all around the world
- Around 450,000 servers
 - Estimate 2015: over 1,000,000 servers
- 200 petabytes of hard disk storage – enough to copy the entire Net dozens of times –
 - *Petabyte* = 10^{15} bytes
- 4 petabytes of RAM
- Receives 100 million queries a day
 - 2016: 2.3 million queries/second
- Collective input-output bandwidth around 3 petabits per second

Drawbacks of web search engines

- User has to specify a **concrete query**
 - They typically return a **huge number of hits**
 - Problems of **natural language ambiguity**
 - The results are not structured in any way (**list**)
 - The search engine does **not learn** anything about the user
 - ❑ Not adaptive
 - ❑ Same behaviour for each user
-

Searching for information on the Web

■ Option 2: Information agents

- ❑ Learn (automatically) the preferences of the user => **user model, user profile**
 - By supervising the user activities
 - By receiving explicit information/instructions/feedback from the user
- ❑ Adapt the behaviour to the user preferences

Example: Letizia

Heterogeneous systems

- **Homogeneous** system: all agents belong to the same type (e.g. a group of collaborative agents)
 - **Heterogeneous** multi-agent system
 - Set of 2 or more agents that belong to 2 or more different types
-

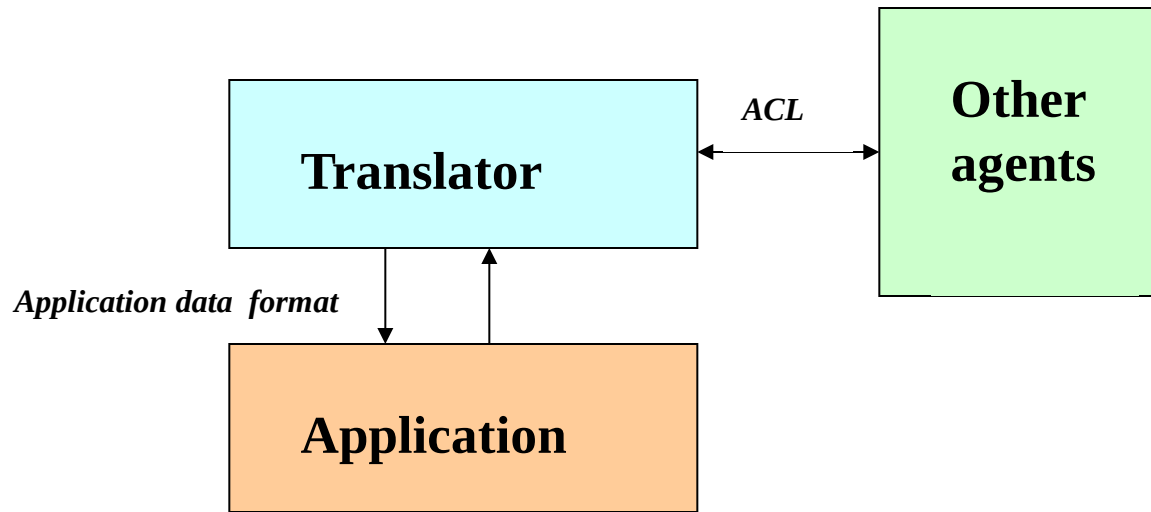
Heterogeneous system: Rationale

- **Join** different existing programs/applications (not necessarily based on agent technology) in a single agent-based system
 - *Legacy* systems
 - **No need to re-program** all the applications to integrate them within a MAS
 - The union can give an **added value**
-

Agentification

- Ways to transform a standard application into an “*agent*” that can be integrated and participate in a multi-agent system of collaborative agents to help to solve complex distributed tasks
-

Option 1: Translator / Transductor

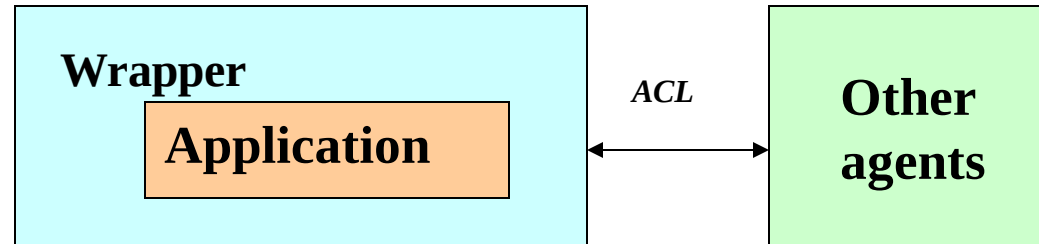


- “**Bridge**” between the application and the other agents
- Takes the messages of other agents, and translates them to the program communication protocol (and the other way round)

Advantages

- The code of the application does not have to be modified
 - We only need to know the **inputs/outputs** of the application
 - The same translator can be used to agentify different applications/resources
 - Access to an Oracle database
 - Access to a CLIPS rule-based system
-

Option 2: Wrapper



- **Add code** to the application so that it can communicate with the agents of the MAS
 - Implies a direct manipulation of the internal data structures of the application

Comments on wrappers

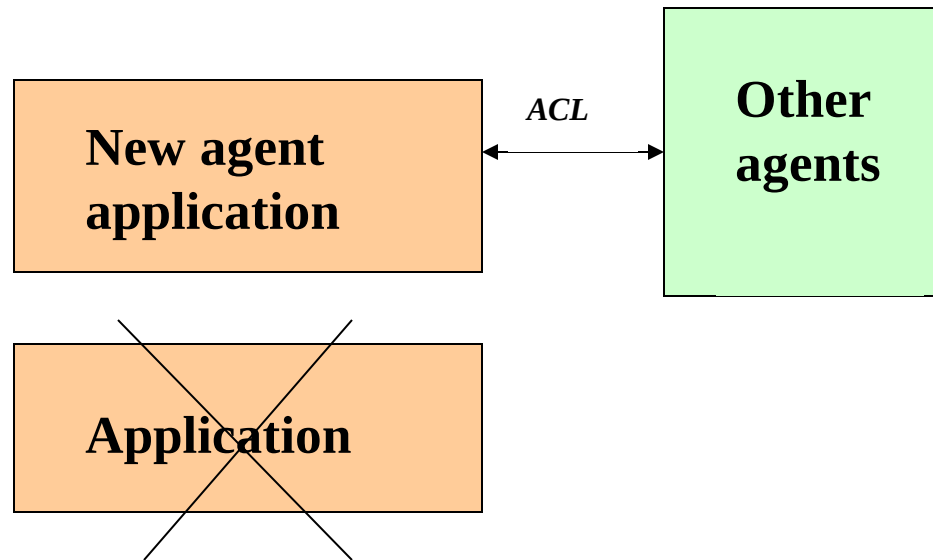
- Positive aspects

- More computationally **efficient** than translators

- Negative aspects

- We need access to the application **source code**
 - It may **not be easy** to understand/modify the code of a complex application
 - A wrapper is **not reusable** as a translator
 - One wrapper for each program to be agentified
-

Option 3: Re-code the application



- **Re-write** the original program as an agent

Comments on re-writing

- **Negative** aspects

- Much more **work** than the previous options
- It does **not** seem really an “**agentifying**” mechanism

- **Positive** aspect

- The resulting agent could be designed to work in a much more **efficient** way, without external/internal translations between formats
-

Architectures for heterogeneous systems

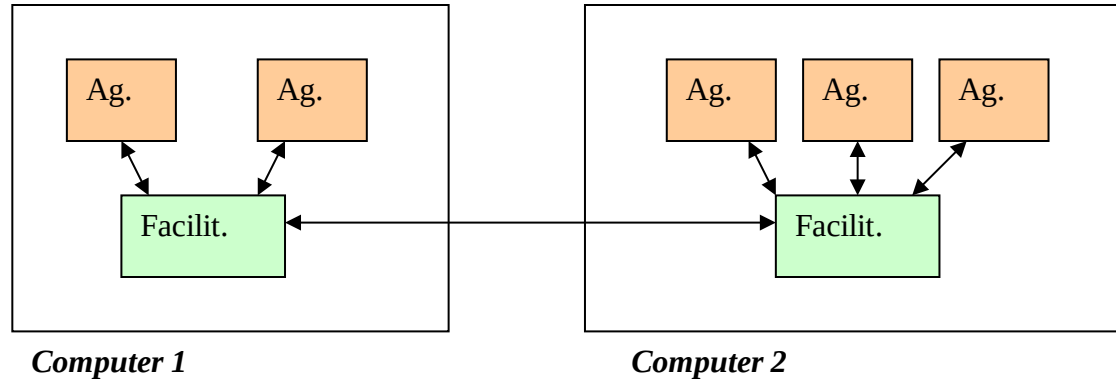
- **Flat** system

- Each agent can talk directly to any other agent

- **Federated** system

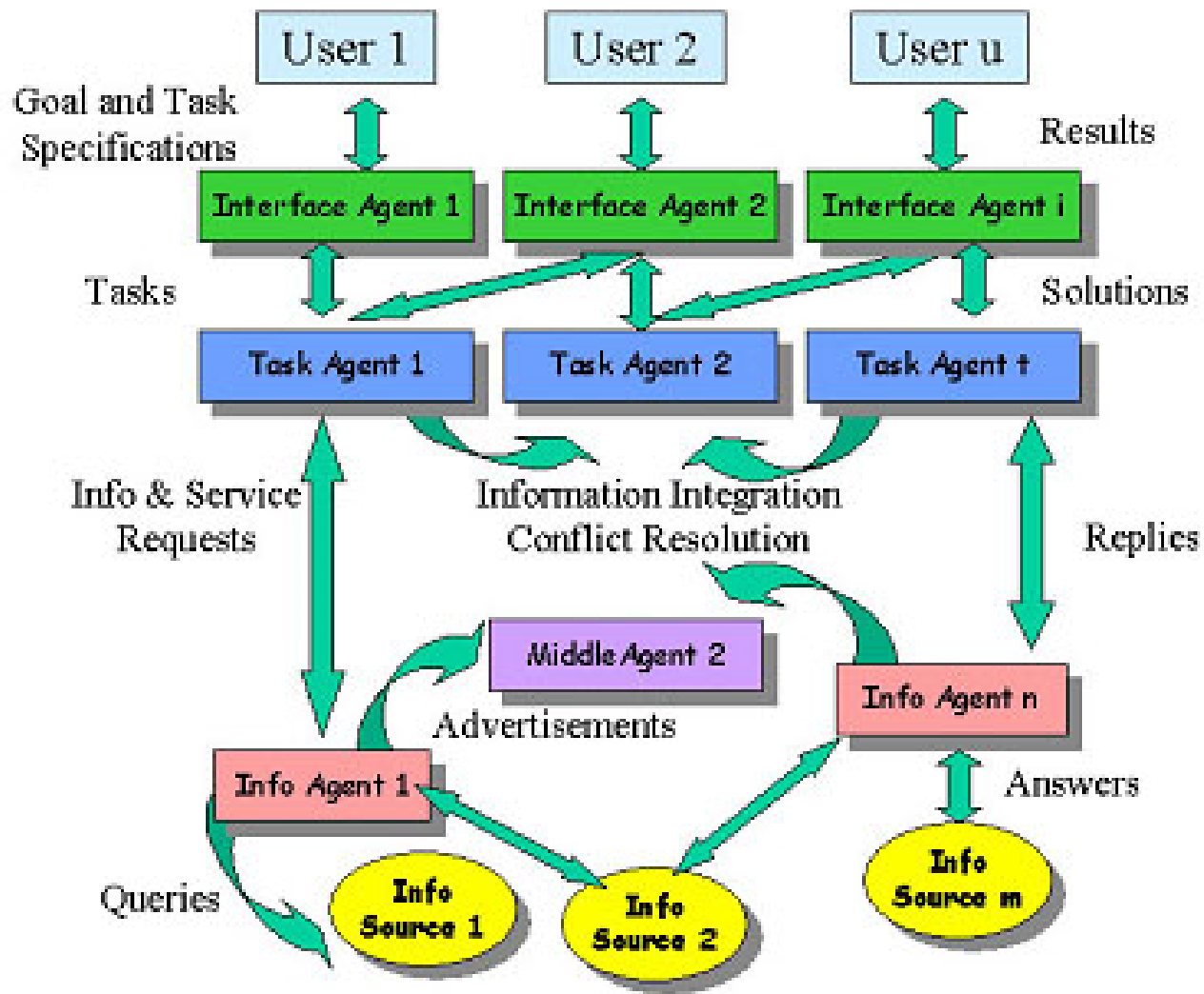
- There are special agents (**facilitators**), that manage
 - the connection of the system with the users, and
 - the communication between the agents

Federated systems



- Agents do not communicate directly, but through **facilitators**
- Facilitators help agents to find other agents in the system that provide some services
 - Without having to ask all other agents

Example: RETSINA



RETSINA agents

- **Interface agents**: input/output, learn from user actions
 - **Task agents**: encapsulate task-specific knowledge, used to perform/request services to/from other agents/humans
 - They can coordinate their activities
 - **Middle agents**: infrastructure service (e.g. they know the services provided by other agents)
 - **Information agents**: monitor and access one or more information sources
-

Final comment

- This classification is not intended to be either exhaustive or a very strict division of agents
 - Mobile information agents
 - Collaborative interface agents
 - Collaborative heterogeneous system
 - An agent can fall, to a certain degree, in different categories
-

Readings for this week

- Article on Software Agents (Nwana, BT Lab)
 - Link to *Letizia* at MIT
 - Klusch: *Information agent technology for the Internet: a survey*
 - Wired (Oct06) article: *The information factories*
 - Extra information on RETSINA
 - Other material
-

LECTURE 3:

AGENT TYPES

Introduction to Multi-Agent Systems (MESIIA, MIA)

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